

Multiple Imputation Strategy for Overall Survival (OS) Outcome

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Overview

This document describes the multiple imputation strategy for the Overall Survival (OS) outcome in the LIQPLAT trial. The approach uses Multiple Imputation by Chained Equations (MICE) to handle missing baseline covariate data under the Missing At Random (MAR) assumption.

Since OS data are structured in wide format (one row per patient), we use standard single-level imputation methods appropriate for each variable type.

Setup

```
library(tidyverse)
library(here)
library(mice)
library(arrow)
```

Data

The following code uses historical data from the Division of Medical Oncology at USB (first appointments from 2023 to mid-2024, prior to LIQPLAT initiation). These data are used to develop and validate the imputation procedure; actual trial data are not publicly available.

```
data_raw <- read_parquet(here("data", "wide_data_730.parquet"))
```

Data Preprocessing

Variable Transformations

Survival time is censored at 182 days (6 months) for the primary analysis. The death event indicator is derived accordingly. Diagnosis categories are collapsed to the 5 most frequent groups to avoid sparse categories that could cause convergence issues.

```
data <- data_raw |>
  mutate(
    diagnosis = fct_lump_n(diagnosis, n = 4),
    diagnosis = factor(as.integer(diagnosis)),
    event_death = if_else(survival_time <= 182, 1, 0),
    survival_time = pmin(survival_time, 182),
    tx = factor(if_else(randomselect_liqplat == "1", 1, 0)),
    ecog_fstcnt = factor(ecog_fstcnt, ordered = TRUE)
  ) |>
  select(
    pat_id,
    gender,
    age,
    ecog_fstcnt,
    tx,
    albumin,
    bilirubin,
    c_reactive_protein,
    lactate_dehydrogenase,
    event_death,
    survival_time,
    diagnosis
  ) |>
  relocate(pat_id, event_death, survival_time)
```

Nelson-Aalen Cumulative Hazard Estimate

We include the Nelson-Aalen cumulative hazard estimate (na_est) as a predictor in the imputation model. This is the recommended approach for incorporating survival information into imputation models, as it provides a smooth summary of the survival process that can predict missing baseline covariates [van-Buuren2021-pc, chapter 9.1].

```
data$na_est <- nelsonaalen(data, timevar = survival_time, statusvar = event_death)
```

Imputation Variables

The imputation model includes variables from the substantive analysis model and auxiliary variables that are predictive of missing values or their missingness mechanism.

Analysis Variables

Variable	Description	Type
tx	Treatment group allocation	Binary
ecog_fstcnt	ECOG performance status at baseline (0–4)	Ordered factor
diagnosis	Primary cancer diagnosis (5 categories)	Nominal factor
albumin	Serum albumin at baseline (g/L)	Continuous
c_reactive_protein	C-reactive protein at baseline (mg/L)	Continuous
event_death	Death indicator (within 182 days)	Binary

Table 1: Analysis variables included in the imputation model

Auxiliary Variables

Variable	Description	Type
age	Age at date of randomisation (years)	Continuous
gender	Patient sex	Binary
bilirubin	Serum bilirubin at baseline ($\mu\text{mol/L}$)	Continuous
lactate_dehydrogenase	Lactate dehydrogenase at baseline (U/L)	Continuous
na_est	Nelson-Aalen cumulative hazard estimate	Continuous

Table 2: Auxiliary variables included in the imputation model

Multiple Imputation Procedure

Since OS data are in wide format (one observation per patient), we use standard single-level MICE methods appropriate for each variable type.

Imputation Methods

Variable	Method	Description
ecog_fstcnt	polr	Proportional odds logistic regression for ordered factors
diagnosis	polyreg	Polytomous logistic regression for un-ordered factors
albumin	pmm	Predictive mean matching for continuous variables
bilirubin	pmm	Predictive mean matching for continuous variables
c_reactive_protein	pmm	Predictive mean matching for continuous variables
lactate_dehydrogenase	pmm	Predictive mean matching for continuous variables

Table 3: Imputation methods by variable

Predictive mean matching (PMM) is used for continuous variables to preserve the marginal distributions and avoid imputing implausible values. We use Type 1 PMM with a donor pool of $k = 5$ [van-Buuren2021-pc,

chapter 3.4]. For ordered categorical variables (ECOG), proportional odds logistic regression (`polr`) maintains the ordinal structure. For unordered categorical variables (diagnosis), polytomous logistic regression (`polyreg`) is used.

Predictor Matrix Configuration

The predictor matrix is configured to:

- Exclude `pat_id` from predictions (identifier only)
- Exclude `survival_time` from predictions (we use `na_est` instead to incorporate survival information)
- Allow all other variables to predict each other

```
md.pattern(data)
```

```
init <- mice(data, maxit = 0)
meth <- init$method
pred_matrix <- init$predictorMatrix

# Patient ID should not be used for imputation
pred_matrix[, "pat_id"] <- 0
pred_matrix["pat_id", ] <- 0
meth["pat_id"] <- ""

# Use Nelson-Aalen estimate instead of raw survival time
pred_matrix[, "survival_time"] <- 0
pred_matrix["survival_time", ] <- 0
meth["survival_time"] <- ""

imputed_data <- mice(data,
  m = 10,      # 50 in final analysis
  maxit = 10,  # 50 in final analysis
  predictorMatrix = pred_matrix,
  method = meth,
  donors = 5,
  seed = 1234)
```

Imputation Parameters

For the final trial analysis:

- Number of imputed datasets: $m = 50$
- Maximum iterations: 50
- Donor pool size: $k = 5$

Convergence Diagnostics

Convergence is assessed via trace plots of the mean and standard deviation of imputed values across iterations.

```
plot(imputed_data)
```

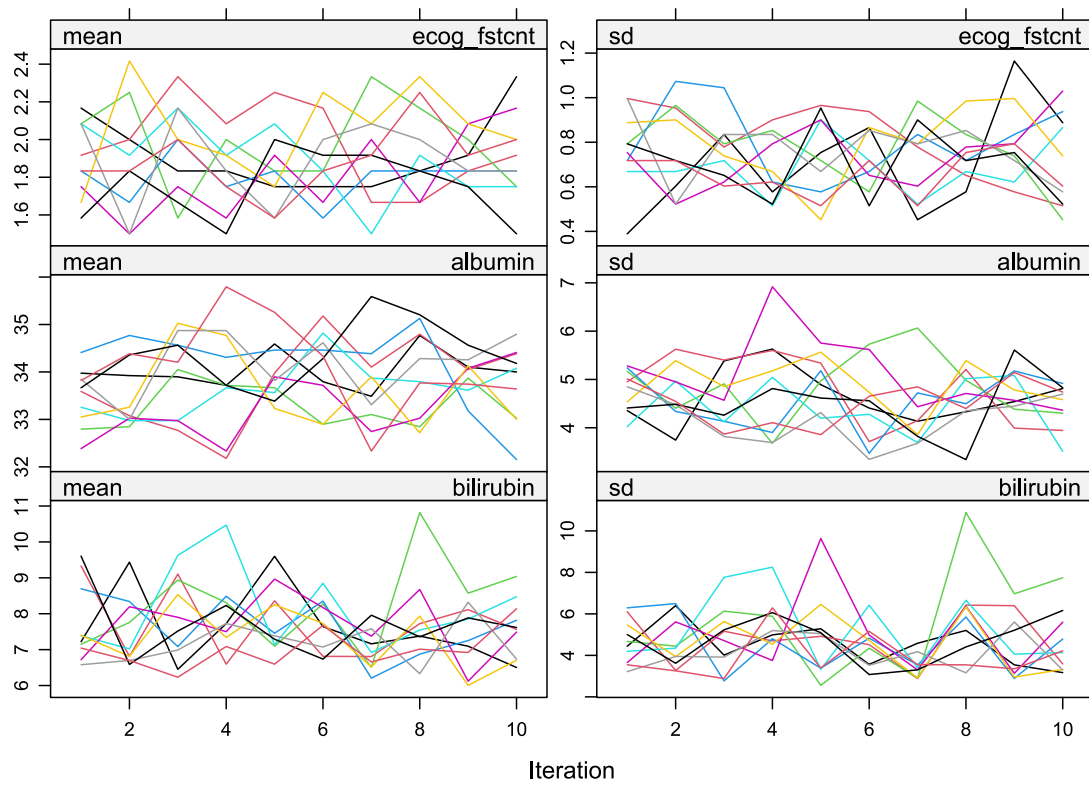


Figure 1: Trace plots showing the mean and standard deviation of imputed values across iterations for each imputed dataset.

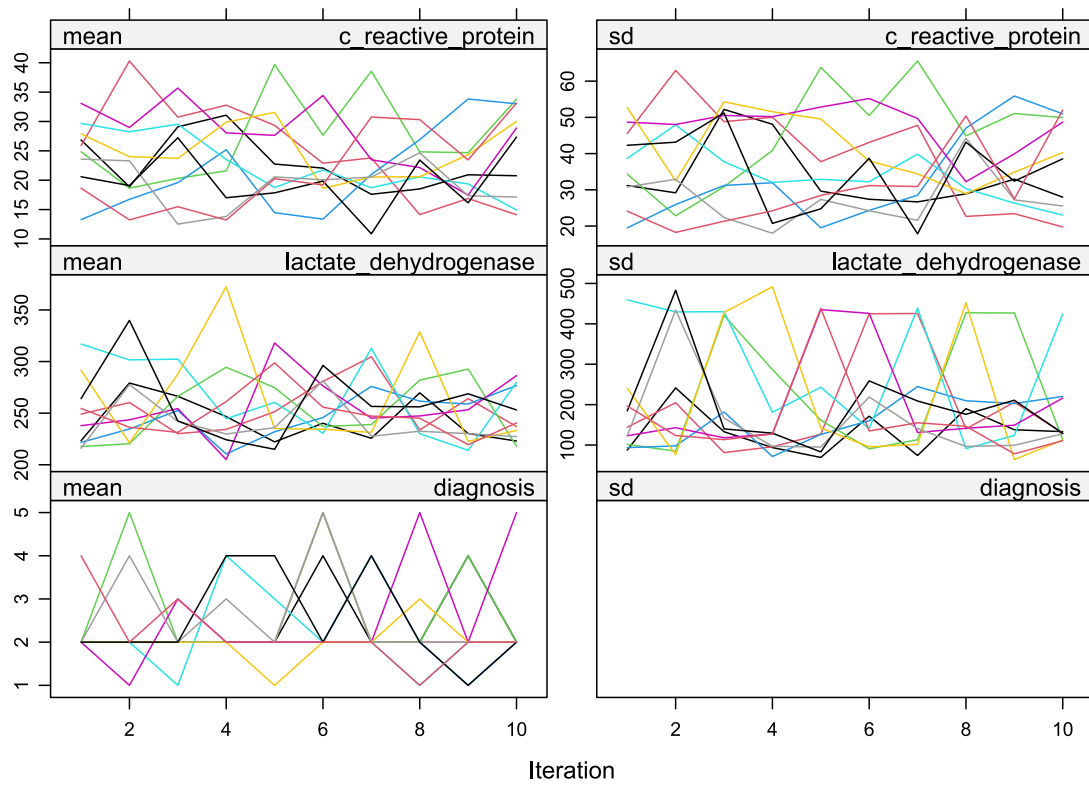


Figure 2: Trace plots showing the mean and standard deviation of imputed values across iterations for each imputed dataset.

Distribution Diagnostics

We compare the distributions of observed and imputed values to assess plausibility.

```
densityplot(imputed_data)
```

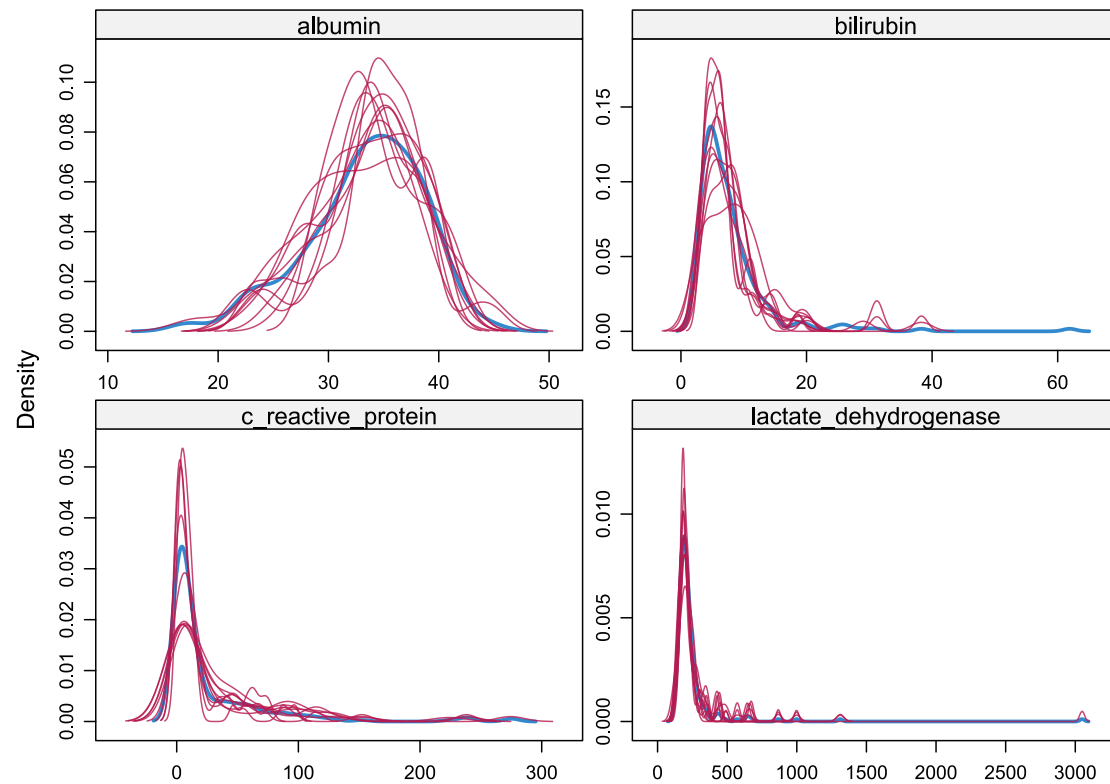


Figure 3: Density plots comparing observed (blue) and imputed (red) values for all imputed variables.

```
densityplot(imputed_data, ~ecog_fstcnt)
```

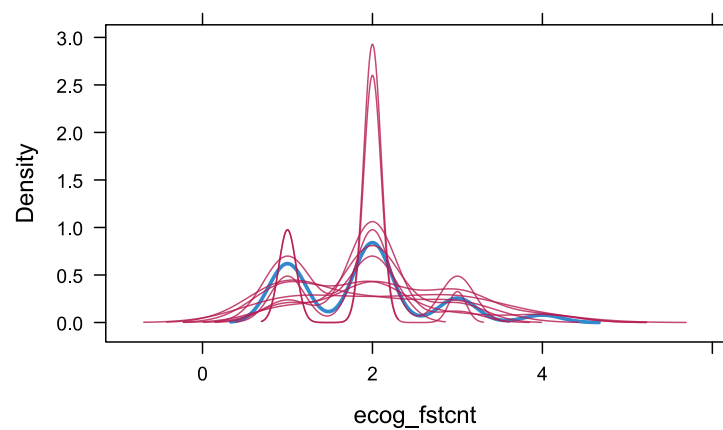


Figure 4: Density plot for ECOG performance status.

```
stripplot(imputed_data, albumin + bilirubin + c_reactive_protein +  
lactate_dehydrogenase ~ .imp)
```

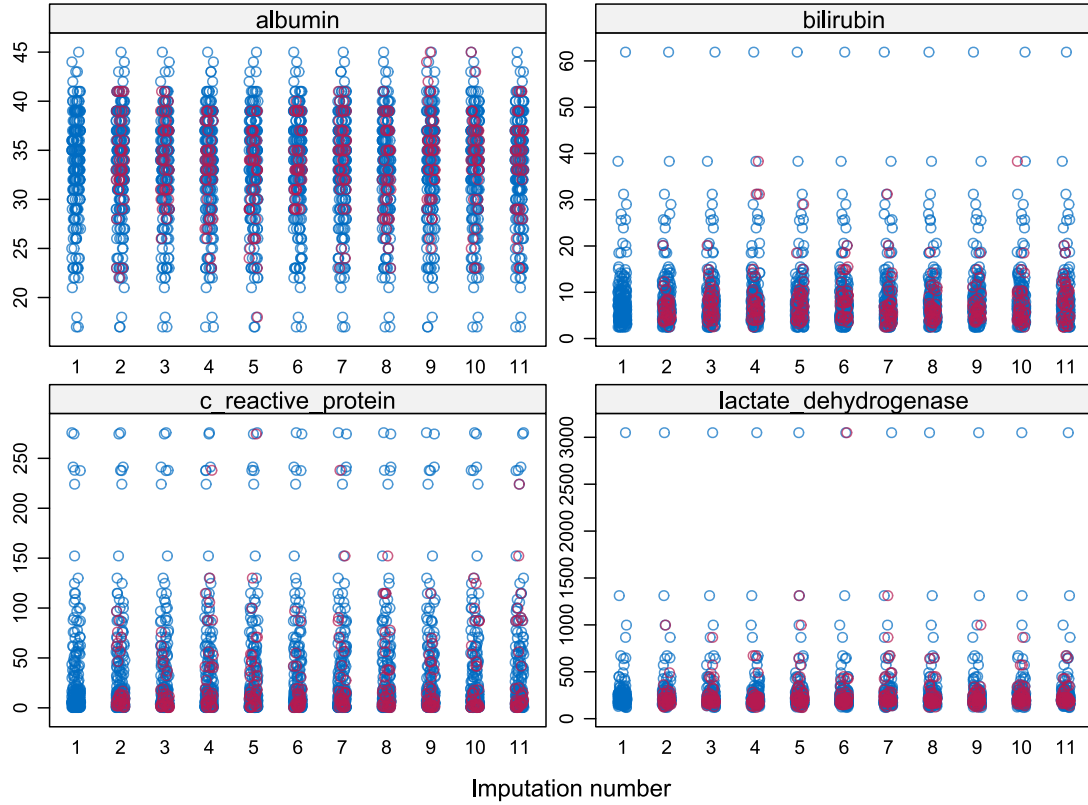


Figure 5: Strip plots of imputed values for continuous laboratory variables across imputed datasets.

```
stripplot(imputed_data, ecog_fstcnt ~ .imp)
```

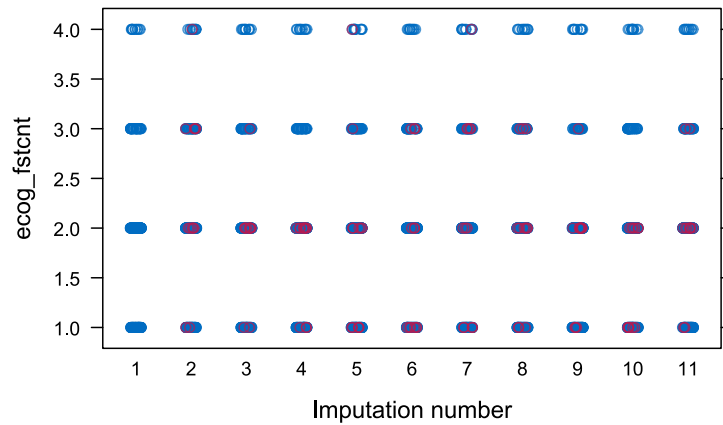


Figure 6: Strip plot of imputed values for ECOG performance status across imputed datasets.

Analysis of Trial Data

For the final trial analysis, all $m = 50$ imputed datasets will be analysed, and posterior draws will be stacked to propagate imputation uncertainty into the final inference.

Simulation Study Dataset

For simulation studies, a single completed dataset is extracted. This is computationally necessary but does not reflect the full imputation uncertainty.

```
complete_data <- complete(imputed_data, 4)
```

Export

```
write_parquet(complete_data, sink = here("data", "survival_complete.parquet"))
```